# Standard library and typing cheat sheet

Python Lab cheat sheet. Snapshot 17 October 2026. This is a study aid, not a specification. Confirm details in the Sources section of Python Lab and on the official pages linked there.

## Reach for these first

- pathlib for paths. open, or Path.read_text, with an explicit encoding for text.
- json, csv, argparse, datetime, statistics, math, random, re.
- collections (Counter, deque, defaultdict), itertools, functools, heapq.
- sqlite3 for a local database file. unittest for tests that ship with Python.
- venv creates virtual environments. pip installs packages into the active one.
- typing holds annotation forms. The interpreter does not type-check your program.
- asyncio runs coroutines on one thread. await yields. It does not make CPU work parallel by itself.

## Annotations you will see

- list[int], dict[str, int], tuple[int, ...], set[str].
- X | None is the modern optional form. Optional[X] still exists.
- TypedDict, Protocol, and TypeAlias describe shapes. They are not runtime schemas unless a library uses them that way.
- A type checker such as mypy or Pyright is a separate tool.

## Packaging

- pyproject.toml is the project file. PEP 621 describes the [project] table.
- A wheel is a built distribution. An sdist is a source distribution.
- uv and ruff are third-party tools from Astral. They are not the standard library.
- pytest is third-party. unittest is the standard-library test runner.